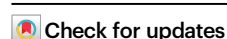


Integrating artificial intelligence with mechanistic epidemiological modeling: a scoping review of opportunities and challenges

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Integrating prior epidemiological knowledge embedded within mechanistic models with the data-mining capabilities of artificial intelligence (AI) offers transformative potential for epidemiological modeling. While the fusion of AI and traditional mechanistic approaches is rapidly advancing, efforts remain fragmented. This scoping review provides a comprehensive overview of emerging integrated models applied across the spectrum of infectious diseases. Through systematic search strategies, we identified 245 eligible studies from 15,460 records. Our review highlights the practical value of integrated models, including advances in disease forecasting, model parameterization, and calibration. However, key research gaps remain. These include the need for better incorporation of realistic decision-making considerations, expanded exploration of diverse datasets, and further investigation into biological and socio-behavioral mechanisms. Addressing these gaps will unlock the synergistic potential of AI and mechanistic modeling to enhance understanding of disease dynamics and support more effective public health planning and response.

Epidemiological modeling is a powerful tool for understanding the dynamics of infectious diseases and guiding public health decisions and policies^{1–5}. Mechanistic models, grounded in the known governing laws and physical principles of disease transmission, have been widely used to investigate various infectious diseases, including respiratory infections^{6–8}, sexually transmitted diseases^{9,10}, and vector-borne diseases^{11,12}. Unlike empirical models, which primarily focus on data fitting without necessarily incorporating the underlying causes of observed patterns, mechanistic models aim to explain how and why epidemics unfold.

Despite their utility for predicting and controlling the spread of infectious diseases, traditional mechanistic models, such as the classical susceptible-infected-recovered (SIR) structure, face several challenges. First, the reliability of these models depends heavily on the accuracy of estimated parameters governing transmission dynamics^{5,13,14}. However, current models are often constrained by simplifications and data availability. For example, disease transmissibility, though modeled as dynamic, is frequently calibrated using lagged and potentially incomplete death or hospitalization data. Similarly, human contact patterns, crucial for understanding

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